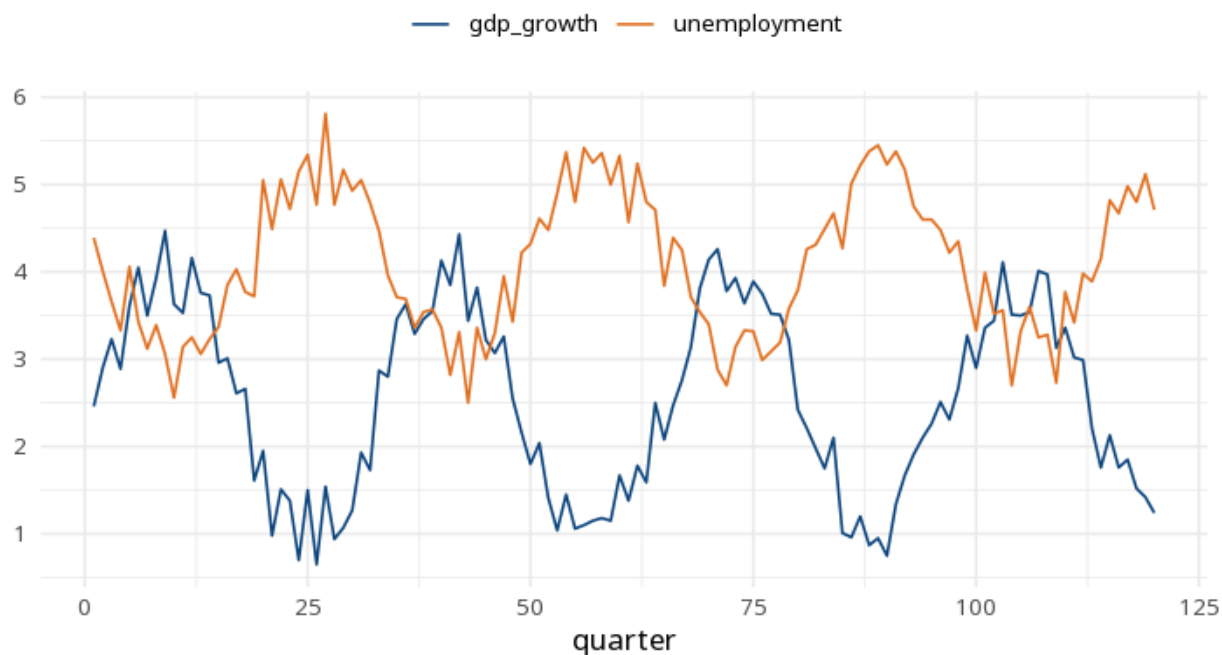


1 Granger causality

Direction	F	df	p
X→Y	122.15	4, 111	<.001
Y→X	3.42	4, 111	.011

Lag	AIC	BIC	HQ
1	-5.14	-5.00	-5.08
2	-5.14	-4.90	-5.04
3	-5.17	-4.84	-5.04
4	-5.23	-4.81	-5.06



A significant $X \rightarrow Y$ p means past values of X help predict Y beyond Y's own past - this is predictive precedence, not proof that X causes Y. Check both directions. The result depends entirely on the lag order (this card uses a max lag of 4): choose it on theory or by an information criterion - vars::VARselect reports the AIC- and BIC-minimising lag - and report it, since different lags can flip the conclusion. Make both series stationary first. Method: R lmtest::grangertest (Granger, 1969). Your significance threshold (α) is 0.05.

Granger test $X \rightarrow Y$: $F(4,111)=122.15$, $p < .001$; $Y \rightarrow X$: $F(4,111)=3.42$, $p = .011$ (lag=4).

Statistical basis: Granger causality F-test (Granger, C. W. J. (1969). "Investigating causal relations by econometric models and cross-spectral methods."

Econometrica, 37(3), 424-438.); Implementation (`lmtest::grangertest`) (Zeileis A, Hothorn T (2002). "Diagnostic Checking in Regression Relationships." R News, 2(3), 7-10.).